When Is Video Enough? Predicting Deformable Object Responses to New Interactions

Supplementary Material

Anonymous CVPR submission

1. Experimental settings

The dynamic stem experiment measures the response to a calibrated load, once with noisy supplied tracks and once with tracks extracted from RGB-D. A separate compliant-base stem tests the decision to proceed, obtain a new view or apply a probe. Torus, drop and bowl experiments examine temporal extrapolation, mechanical ambiguity and numerical mismatch. The RGB-only three-interaction protocol describes the next comparison separately from completed measurements.

The joint-field experiments later in this supplement use known geometry, calibrated cameras and supplied surface correspondences. They assess how uncertainty in applied loads, supports and material fields affects quasistatic displacements. The original observations span the loading history, so unmeasured times test interpolation. Separate reset loads test prediction under changed interactions.

2. Dynamic stem under calibrated forces

A scale ambiguity with an external test. The stem has a 0.08×0.08 m square cross-section and free length $L = 0.40$ m. A clamp fixes the first 0.04 m of its 0.44 m total length. The three material settings pair densities $\rho = 500, 1000, 2000$ kg/m^3 with Young's moduli $E = 2, 4, 8$ MPa, respectively. All have Poisson ratio 0.3. For motion driven by gravity from the same undeformed, stationary initial state, mass, gravitational loading and elastic restoring forces scale together. Dividing the equations of motion by ρ leaves the common ratio E/ρ. The nonlinear reference simulations agree to within 4×10^{-17} m during passive release. A prescribed force in newtons breaks this symmetry.

At 48 frames/s we observe a passive release (D0, 0.5 s) and optionally a second release with a 40 N lateral load distributed around $0.50L$ (D1, 0.75 s). The force rises with a 0.25 s cosine ramp. D2 applies 20 N laterally at $0.75L$. The fresh diagonal test D2b applies 15 N downward and 20 N laterally at the same station. For the D2 lateral target, an estimator that predicts one peak for both extreme worlds must miss at least one peak by $(30.7 - 7.7)/2 = 11.5$ mm to the reported precision. This bound concerns peak displacement at $0.75L$. We score displacement trajectories separately.

Reference and visual baseline. The reference integrates compressible Neo-Hookean dynamics using quadratic tetrahedra on a $33 \times 6 \times 6$ cell grid, implicit Newmark time integration and Newton iteration. On the largest calibration response, a coarser mesh changed station displacement by 0.16 mm RMS and halving the time step changed it by 0.07 mm, compared with 25.7 mm of motion. Our lower-resolution linear-elastic model uses the same element family. GIC reconstructs its particles from RGB. Its material-point simulator fits one global E, Poisson ratio and initial velocity with density fixed at 1000 kg/m^3. The calibrated-force adapter adds the specified momentum to its grid and propagates its derivative. A zero-force replay agreed with native simulation within GPU run-to-run variability, while a momentum check gave a 0.18% difference. GIC refits on the calibration episode from its initial estimate, whereas our posterior multiplies the likelihoods of the two episodes.

Measurements and update. The first study supplied eight 3D material-point tracks with 1 mm noise to our model. The second extracted tracks of 16 known surface markers from the same calibration RGB frames available to GIC. Markers start at their projected rest positions and pyramidal Lucas–Kanade follows their texture in cameras 1 and 9. We back-project each tracked pixel using the median of a 3×3 depth neighborhood after adding independent 1.5 mm Gaussian depth noise. All 16 tracks survived. Their 3D errors had median 0.5 mm and 95th percentile 1.3 mm. This is an RGB-D experiment for our model and an RGB experiment for GIC. Camera 0 is withheld throughout fitting.

Each stem hypothesis specifies $(E/\rho, \nu, \rho)$. Its predicted marker displacements on D0 and D1 are compared with the observations using a Gaussian residual model. We estimate the coordinate noise scale once from the best D0 residual (0.43–0.44 mm in the RGB-D study). The likelihood treats correlated tracking errors as independent. We use the weighted mean trajectory for prediction, and sum the weights of trajectories farther than 3 mm RMS from that

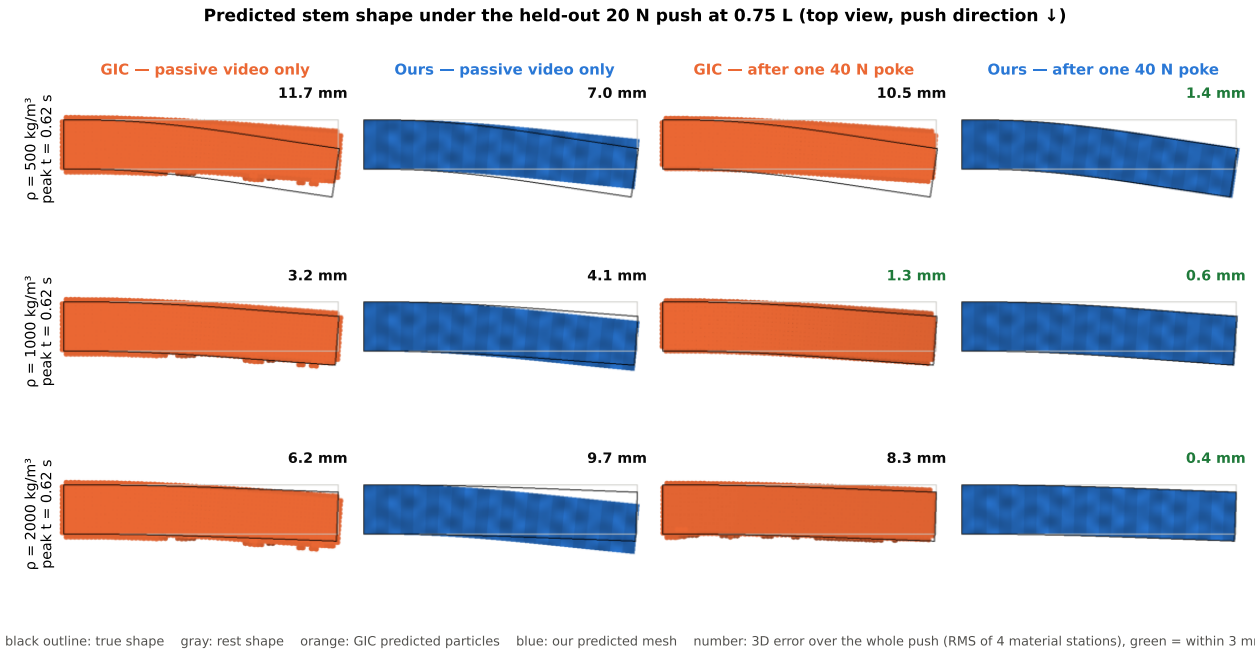


Figure 1. Top views at each reference stem's peak lateral tip displacement under the 20 N test push (D2). Gray and black outlines denote the rest shape and the nonlinear reference, respectively. Orange shows the GIC particle reconstruction from RGB. Blue shows the posterior-mean FEM field inferred from eight supplied 3D tracks with 1 mm coordinate noise, before and after the 40 N calibration. Displacements are drawn at their physical scale. Each annotation gives the 3D four-station error over the entire trajectory. The panels are geometry visualizations and use a different camera from the appearance evaluation.

Table 1. Prospective diagonal push D2b. Each column is one material setting. Errors are 3D displacement RMS in mm. The GIC oracle uses the reference material parameters with GIC's reconstructed geometry.

Density (kg/m^3)	500	1000	2000
Frozen shape	34.0	21.6	15.9
GIC, passive RGB	14.33	3.08	7.18
GIC, RGB after calibration	12.98	1.88	8.08
GIC, oracle material	9.3	7.6	7.1
Ours, passive RGB-D	8.74	5.30	12.08
Ours, RGB-D after calibration	2.75	1.12	0.65
Ours, supplied tracks after calibration	2.75	1.12	0.65
Ours, off-grid, supplied tracks	2.73	1.15	0.71

mean to estimate failure risk. The reported error averages squared 3D displacement errors across four stations ($0.25L$, $0.50L$, $0.75L$, and the tip) and all 37 target frames before taking the square root. Subtracting each method's own rest state removes constant reconstruction offsets.

Outcomes and interpretation. On D2 the post-calibration supplied-track predictor reached 1.4, 0.6 and 0.4 mm RMS from low to high density. After calibration its posterior mode and posterior mean gave the same point accuracy. Before calibration the mode has an arbitrary density tie because D0 is scale-invariant. For D2b the motion predictions and fitted appearance checkpoints were frozen before generating the reference response (Tab. 1). The estimated RGB-D risk for the 3 mm tolerance was about 0.90 in all three worlds before calibration and zero in all three after it. The largest residual, 2.75 mm, approached that tolerance. The three cases alone do not evaluate probability calibration across a range of risks. The GIC oracle retains 7–9 mm RMS. Its particle reconstruction overestimates stem volume by 29%, and its bending response differs from the FEM reference. Both sensing and simulator differences contribute to this comparison.

What changed in the rendering analysis. The original analysis plan specified rendering our field on the known textured rest mesh and averaging PSNR over frames. The reported comparison instead drives the same frozen GIC appearance with the competing motion fields. For each noninitial frame we compute mean squared RGB error e_t on the reference foreground of camera 0, after black-background compositing. For unit-range RGB the reported score is $-10 \log_{10}(T^{-1} \sum_t e_t)$. This gives equal weight to each of the 36 noninitial frames of D2b before conversion to decibels. These analysis choices were amended, while the D2b

physical predictions remained sealed. Applying the RGB-D method retrospectively to D2 also uses a target that had already been evaluated.

With GIC's D0 appearance and reference motion, D2b scores were 21.49, 21.60 and 21.66 dB. The RGB-D posterior mean after calibration scored 21.27, 21.71 and 21.75 dB through that appearance. The reference-motion values are useful comparison points. Imperfections in appearance allow perturbed geometry to yield a slightly higher image score. GIC's D1 reconstruction has its own learned appearance and particles, so its D2b scores of 17.7, 21.1 and 19.0 dB should be interpreted with its corresponding reference-motion values of 18.5, 21.4 and 20.4 dB. During D0 the GIC deformation network achieves 24.9 dB and its physics replay achieves 20.6 dB. The two modes represent different predictions.

3. Choosing an observation under a limited budget

The 96-case experiment uses a rotation-constrained stem on a translating compliant base. It is distinct from the fixed-clamp dynamic stem above. Let $s = 1/k_b$ be base compliance, EI bending rigidity, and $b = L^3/(3EI)$ tip bending compliance. A known tip force F gives the measured tip displacement $F(b + s)$. Thus exchanging compliance between stem and base preserves the observation. Under a force at a, the displacement at x instead obeys

$$u(x; a, F) = F\left[s + \frac{\min(x, a)^2(3\max(x, a) - \min(x, a))}{6EI}\right]. \tag{1}$$

For example, $(b, s) = (0.4, 0.1)$ and $(0.1, 0.4)$ m/N both yield 10 mm at the tip under 20 mN. Moving that force to the midpoint yields 3.00 and 8.25 mm there. Alternatively, viewing the midpoint under the original tip force reveals 4.50 and 8.625 mm. The view and the probe therefore provide different measurements.

We draw b and s independently from $[0.04, 0.48]$ m/N and approximate the posterior on a 129×129 midpoint grid. Initial and additional view noise has standard deviation 0.10 mm and probe noise is 0.15 mm. A new view can see base motion in one access condition and can only repeat the original tip measurement in the concealed condition. Each probe resets the object and applies a load at $0.35L$, $0.50L$ or $0.65L$. The new target load acts at $0.75L$. The repeat condition uses the original load as the target. For each candidate measurement the policy predicts its observation distribution and the updated risk of exceeding 0.50 mm target error. In failure-probability units the view costs 0.015, a push costs 0.040, and proceeding costs zero. The three access/task conditions are paired within 96 distinct physical cases. Policies choosing the same action receive the same noisy observation.

The full posterior chose to proceed in all repeat cases, look in 92/96 visible-base cases and push at $0.35L$ in 91/96 concealed-base cases. Its 90% marginal response intervals covered 87.5%, 89.6% and 87.5% of outcomes, respectively. Relative to always pushing halfway, the mean failure-plus-cost difference was -0.02257 (95% case-bootstrap interval $[-0.02347, -0.02186]$). Relative to the cost-aware Gaussian moment approximation, the difference was -0.00033 ($[-0.00085, 0]$). The two risk policies made similarly effective decisions in this cohort.

4. Rendering and contact controls

Torus forecast. GIC and PAC-NeRF fit cameras 1–10 on frames 0–7 of the same synthetic torus sequence. Camera 0 and future frames 8–13 were withheld. Both use a fixed density of 1000 kg/m^3. Because their background conventions differ, we score the same foreground pixels in both image sequences (the minimum reference RGB channel on white is below 0.95). Here we average the individual frame PSNR values. GIC's appearance deformation gave 20.47 dB during fitting and 8.29 dB for the future, compared with 9.50 dB from holding its frame-7 image. Its native physics gave 17.79 and 18.44 dB, respectively, and PAC-NeRF gave 18.29 and 18.01 dB. Mean symmetric nearest-neighbor distance to reference particles in the future was 4.13 mm for GIC and 4.55 mm for PAC-NeRF with both clouds sampled to 48,691 points (subsampling floor 2.54 mm). This sequence tests extrapolation of existing motion.

Drop ambiguity in a second simulator. On three released ReconPhys assets, scaling native mass, spring stiffness and damping from $(1, 300, 1)$ to $(2, 600, 2)$ yields passive movie pairs with PSNR 94.6–100.1 dB across four cameras. The score compares the two simulated worlds. Their responses to a known uniform body force per anchor differ by 0.121–0.219 in the simulator's normalized coordinate units, measured by RMS of 21 centroid coordinates over frames 18–24. The two-world set contains the truth, and the same-evidence point fit also recovers the corrected response in this construction.

Failure of a contact-model risk estimate. A 128 g tapered deformable body falls into a hemispherical bowl of radius 90 mm. Across 16 combinations of modulus and friction, the available observations are four tracked overhead surface points with 0.5 mm coordinate noise. One position-controlled calibration experiment is permitted, either a 25 mm push or a 0.2 rad rotation of the bowl. The latter retains the gravity direction and changes tangential wall motion. A separate 45 mm transverse push tests 64 material points at three times. Tool and bowl reactions are withheld. Two noise repetitions per world yield 32 records per reference

discretization. With a matched numerical model, mean MAP motion error falls from 7.59 to 0.023 mm after active selection. A fixed rotation reaches zero error in that finite matched dictionary. Against a finer spatial and temporal reference, mean active error is 3.09 mm. The estimated probability of exceeding 3 mm is 1.86%, compared with 43.75% observed failures. A 50:50 mixture of both discretizations reduces the error to 0.051 mm. This mixture contains the reference discretization by construction. These measurements demonstrate sensitivity to the set of candidate models.

5. Design of the additional RGB-only benchmark

The following three-interaction protocol has been specified for a common visual comparison. Its three-by-three score matrix is pending. The completed numerical results above retain their original sensing and calibration conditions.

GIC and PAC-NeRF receive the same ten fitting-camera RGB sequences. Camera 0 is reserved. For our method, known rest geometry, calibrated cameras and marker rest positions initialize two-view tracks. Visible view pairs must subtend at least 20° at the marker. The tracker uses 21-pixel pyramidal Lucas–Kanade windows and triangulates the two image points. It currently requires all selected tracks to survive. Depth images are withheld. The design supplies each method the same calibration command and withholds the target response. GIC and PAC-NeRF use scalar material fits and force/contact adapters. M-PhyGs has no completed image fit in this comparison.

1. **Changed stem force.** After D0 and the optional 40 N midpoint calibration, a new 20 N force at $0.75L$ has components $(0, -16, -12)$ N. Primary score is RMS displacement error over four stations. This changes the direction relative to both D2 and D2b.
2. **Hidden stiff region.** A $0.24 \times 0.10 \times 0.12$ m block has a bonded base and known density 1000 kg/m^3. A uniformly textured surface conceals a band spanning the full depth and upper three quarters of the block. Its modulus is eight times the 60 kPa background value and its center can lie at $x = -0.06, 0, +0.06$ m. A 30 N central press is the initial episode, and the optional 30 N calibration acts at $x = +0.05$ m. The test applies 30 N at $x = -0.07$ m and scores 27 surface points. Our prior includes homogeneous and piecewise-stiff material fields, whereas the configured GIC and PAC-NeRF fits use one global modulus. A comparison with this roster would combine differences in model capacity and inference.
3. **Striker rebound.** The same three stems and calibration episodes as the first scenario are used for a third interaction. A 0.30 kg sphere of 30 mm radius approaches at 1 m/s along $-z$. Each simulator uses a penalty contact law, and the sphere is free to rebound. The primary endpoint is the absolute error in the axial impulse $m(v_{z,\mathrm{end}} - v_{z,0})$, with station displacement as a secondary endpoint.

The three interactions span two geometries. An equal RGB input does not eliminate differences in rest-geometry knowledge, field priors, or solver fidelity. Held-camera images and the physical response should both be evaluated under the same evidence cutoff.

Untested image-space extensions. The RGB-only implementation triangulates each retained marker and scores its three-dimensional displacement. A proposed alternative would project each candidate FEM trajectory into every fitting camera in which the marker is visible. A forward–backward tracking check could downweight uncertain pixel observations and omit missing views. A second extension would score candidate silhouettes through one frozen appearance model, using reserved fitting views. Reserving those observations would prevent treating the same pixels as independent evidence in both the tracking and silhouette terms. Neither extension has a scored comparison here. The completed shared-GIC rendering experiment used appearance only to evaluate a motion field after prediction.

6. Coordinates, units and discretization

Both quasistatic objects have bounding box $[-.06, .06] \times [-.03, .03] \times [0, .06]$ metres. The stepped object removes the upper-right quarter of the longitudinal/vertical bounding rectangle ($x > 0$, $z > .03$ m), leaving 75% of the bounding volume. A regular grid is split into six tetrahedra per cell, consistently oriented. The primary grid uses $(8, 4, 4)$ cells, giving 225 nodes and 768 tetrahedra for the block and 185 nodes and 576 tetrahedra for the stepped block. Fine truth uses $(12, 6, 6)$ cells. Density is 500 kg/m^3, but is not identified and does not enter quasistatic equilibrium. Poisson ratio is 0.30. Gravity is omitted in the incremental reference. A known, nonzero gravity term requires a separate symmetry argument.

For linear tetrahedra, element stiffness is assembled from the exact constant shape gradients and isotropic Lamé coefficients. Elastic stiffness has the expected rigid modes. The only anchoring is the declared positive distributed Gaussian support. We use unregularized Cholesky factorization and count factorization failures.

Surface integrals use three degree-two quadrature points per triangular face. Gaussian traction weights are normalized over the relevant surface and scattered through the face shape functions. Each mode sums to a unit force. Support stiffness uses a lumped nodal diagonal from the same normalized quadrature. The lumped diagonal approximates the consistent surface mass matrix. Normalization preserves total force under refinement but does not guarantee a converged displacement field.

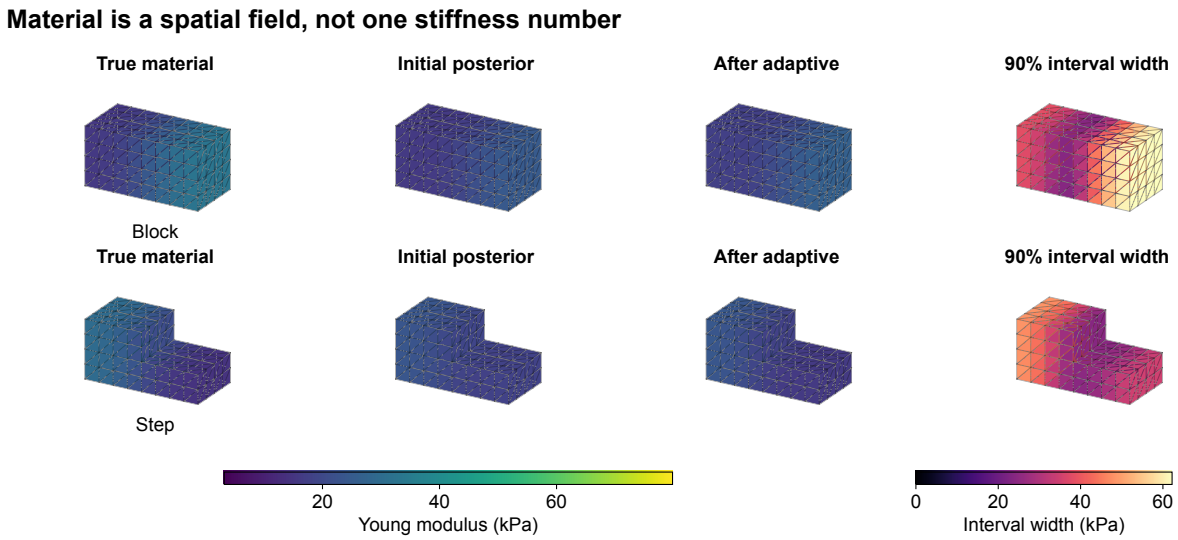


Figure 2. **Material estimates and remaining uncertainty.** Fixed case zero, shared modulus scale and true geometry. These inferred fields can remain uncertain or incorrect despite improving some response predictions. Interval widths are posterior marginal quantities, with cohort coverage evaluated separately.

7. Material and interaction families

Material kernel centers are $(-.035, 0, .03)$ and $(.035, 0, .03)$ metres, with scales $(.035, .06, .06)$ metres. Normalized kernel values define a partition of unity. Their difference is centered by the element-volume average before multiplying log contrast. The reference modulus is the volume-weighted geometric mean of the element moduli.

All support/load footprints have a fixed longitudinal width 0.018 m and are broad in the other surface coordinate. Unknown support location lies in $[-.025, .025]$ m. The unknown top/side load location lies in $[-.04, .04]$ m and is shared between the two modes. The top footprint on the step includes both horizontal terraces. Support stiffness lies in $[4000, 16000]$ N/m, reference modulus in $[8000, 32000]$ Pa, and log contrast in $[-.8, .8]$. The scale priors for material and support are log-uniform. Five equally spaced values in every coordinate define the primary 3125-component dictionary.

The two unknown traction modes act along $-z$ on the top and $-y$ on a side. Their amplitudes may have either sign, so a negative top amplitude represents a pulling actuator. The model assumes signed tractions and bilateral compliant support.

8. Observations and targets

Twenty reference points are declared without using any data. Sixteen span $x \in \{-.045, -.015, .015, .045\}$, $y \in \{-.03, .03\}$ and $z \in \{.015, .025\}$ m. Four additional points use $x \in \{-.045, -.015\}$, both y values and $z = .05$ m. These coordinates lie on the front/back surfaces of both geometries. The primary acquisition and force-scale studies use this surface protocol (v04/v05), as does the temporal follow-up (v06). Earlier v01/v02/v03 reference studies used interior material coordinates at $y = \pm.025$ m and are retained only as mathematical sensor/numerical references. The Cartesian-product ordering fixes all point indices.

Initial measurements use points 1,5,9,13, components x, z, and loading times 0, 3 and 7, for 24 scalar observations. Independent measurement noise has standard deviation 2×10^{-5} m. Two force modes at eight uniformly spaced loading times have a separable Matérn-$3/2$ temporal covariance with correlation length 0.30 and amplitude 0.08 N. The two directions are independent under the prior.

The ten camera candidates comprise six oblique measurements at back-surface points 3 and 11 at times 1, 4 and 6, plus four z-component measurements at front points 1 and 13 at times 2 and 5. For the back view, the outward camera direction is $(-1, 1, 0)/\sqrt{2}$ and measured direction is $(1, 1, 0)/\sqrt{2}$. Initial/front views use camera direction $(0, -1, 0)$. Six calibrated probes use top/side 0.08 N loads at $x = -.04, 0, .04$ m and measure point 13. Top probes measure z from the front. Side probes measure $(1, 1, 0)/\sqrt{2}$ with outward camera direction $(1, -1, 0)/\sqrt{2}$. Thus each measured direction lies in its camera image plane. A deformed-surface visibility audit checks all loading times and all six probe states, using explicit ray–triangle intersection, surface membership and outward normals. It checks 27,776 marker/view instances over all 320 primary and 128 hierarchical cases, with zero failures. This verifies geometric visibility of supplied markers, not correspondence correctness or recoverability from RGB. Each candidate yields one scalar at the same stated noise level. Probe measurements use an unloaded/reset ob-

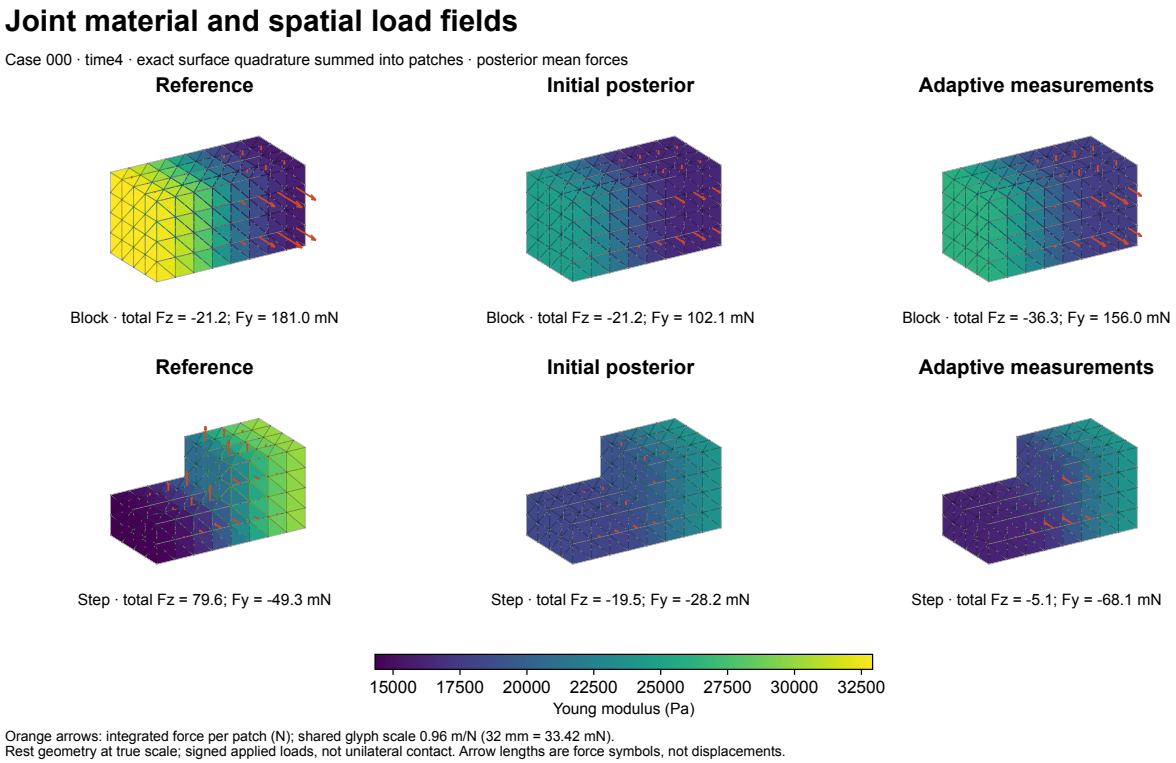


Figure 3. **Joint material and actual surface force distributions.** Fixed case zero, time index 4. Color shows inferred material modulus. Arrows show integrated force per surface patch in newtons, obtained from the exact FEM surface quadrature. All panels share an explicitly stated arrow glyph scale. The stepped object's posterior-mean top-force sign remains incorrect, despite improving response predictions.

ject with unchanged material/support parameters. Original loads are replayable for additional camera observations. The cost model does not equate a physical image with a scalar observation or account for actuator complexity.

The predictive target is fixed before action selection. Eight positive-y points, excluding points 3 and 11, at all eight times give 192 original-motion coordinates. The first sixteen reference points under four reset loads at $x = -.02, +.02$ m, top/side, give 192 changed-load coordinates. None of the original target points is acquired, and none of the target reset loads is used as an acquisition probe. This prevents a policy from appearing better merely because its chosen measurement became its evaluation target.

9. Conditional and total uncertainty

For hypothesis h, write $y = H_h a + d_h + \epsilon$, $a \sim \mathcal{N}(0, \Lambda)$ and $\epsilon \sim \mathcal{N}(0, R)$. The normalized marginal likelihood is

$$\log p(y \mid h) = -\tfrac{1}{2}\{r_h^T C_h^{-1} r_h + \log|C_h| + n\log(2\pi)\}, \quad (2)$$

with $r_h = y - d_h$ and $C_h = R + H_h \Lambda H_h^T$. Every changing determinant is retained. The finite posterior weights multiply this likelihood by the declared hypothesis prior.

For $q = T_h a + b_h$, conditional means and variances are combined by total expectation and variance. The between-hypothesis term is part of the reported prediction and force uncertainty. Marginal equal-tail intervals invert the full Gaussian-mixture CDF. Deterministic material fields are a weighted discrete distribution, whose atoms can produce conservative intervals. Prediction coverage uses 24 fixed coordinates, material coverage 20 fixed points, and force coverage 16 amplitudes. Coverage is evaluated coordinate by coordinate.

The moment-independence ablation preserves the force mean/covariance and the material marginal, but replaces their dependence with one shared moment-matched Gaussian force distribution. Its mean-prediction calculation is a useful dependence control. It approximates the product of the full mixture marginals. Joint-MAP selects the maximum of $\log w_h - \frac{1}{2}\log|S_h|$, whereas marginal-MAP selects $\log w_h$. These are different posterior summaries when the conditional covariance changes.

10. Acquisition calculation and audits

For scalar action s, the component predictive mean is $\mu_{s,h} = h_{s,h} m_h + d_{s,h}$ and variance $v_{s,h} = h_{s,h} S_h h_{s,h}^T + \sigma^2$. Hypothetical observations update Gaussian conditionals and mixture weights. The expected average target variance is estimated with 128 posterior-predictive draws. The same

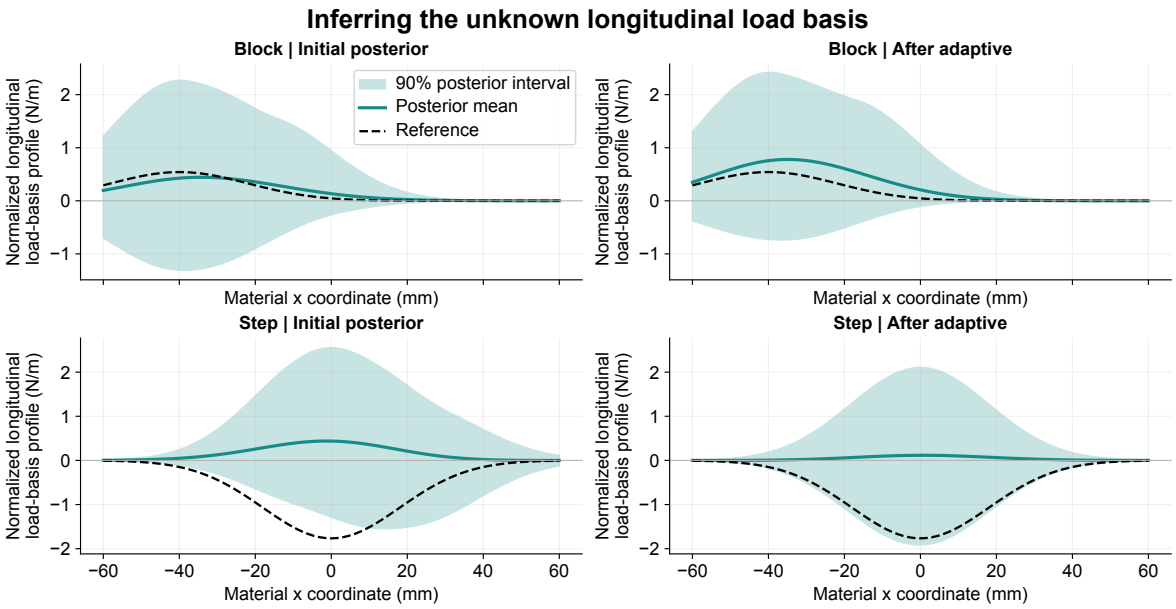


Figure 4. **Unknown spatial loads.** Fixed case zero at the predeclared, initially unobserved time index 4. Normalized longitudinal load-basis profiles and their mixture intervals retain uncertainty in both load amplitudes and their footprint. This profile is a basis proxy: the stepped side surface has varying height, so the one-dimensional profiles and surface-integrated tractions have different normalizations. Ground truth is available only because this is a controlled synthetic experiment.

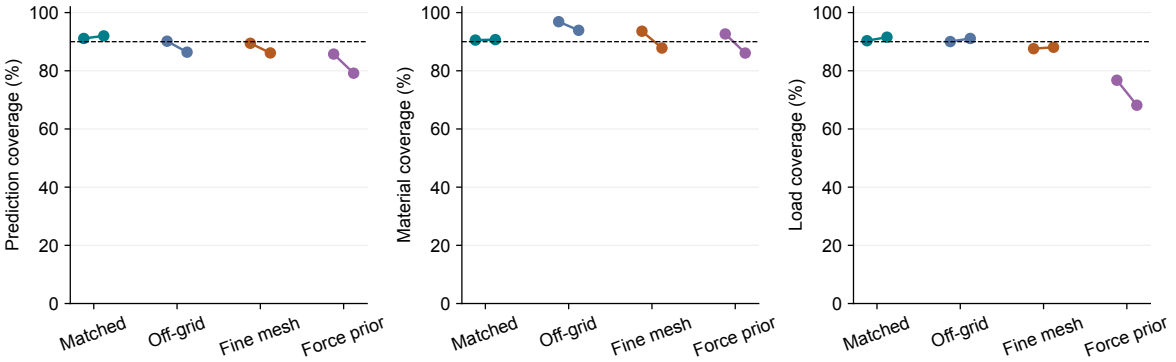


Figure 5. **Complete surface-study marginal coverage.** Initial versus four-adaptive-measurement results, including material and load coverage. Same-model calibration and mismatch robustness are distinct tests.

random numbers are used across candidates. The acquisition RNG has a separate namespace from truth/noise generation. The selected observation alone updates the actual posterior.

Fixed mixed, camera-only and probe-only schedules are greedily selected from twelve separate development cases per shape using their average realized target squared error. These schedules are fixed across the test cohort. Random chooses without replacement. All policies receive the same case-specific potential observation noise, making their comparison paired.

The independent surface audit replays the first four matched cases per geometry, all four decisions, with 1024 predictive draws per candidate. It retains the original actions. 29 of 32 choices agree. Maximum estimated chosen-action regret is 1.529% of current risk. Two disagreements have descriptive paired 95% lower bounds above zero, but selecting the audit minimum introduces bias: these are not selection-corrected optimality tests.

A separate diagnostic halves/doubles the assumed force-prior standard deviation while replaying identical original observations and acquired actions. After acquisition, material means change by 30.0%/22.5% relative to nominal; predictive means change by 0.0185/0.0098 mm. Nominal 90% coverage on these eight cases is 91.67%, compared with 73.96%/92.71% under the changed priors. These small-cohort descriptive fractions quantify sensitivity, not popu-

lation calibration. The separate hierarchy study evaluates uncertainty in amplitude scale on fresh cases.

11. Hierarchical force-scale protocol

A separate fresh cohort samples the same finite spatial prior and one of five equally probable amplitude scales $\tau \in \{.5, 1/\sqrt{2}, 1, \sqrt{2}, 2\}$. There are 64 cases per geometry. Conditional forces retain the same temporal correlation. The resulting 15,625-component posterior is evaluated exactly by reusing mechanical factorizations across scales. Full mixture distributions, rather than a moment-matched collapse, supply intervals and evidence updates.

The hierarchy, a nominal-scale fit, a single variance-matched Gaussian and an oracle given the true scale receive identical observations. The broader Gaussian's standard-deviation multiplier is $\sqrt{1.55} = 1.24499$. Four fixed views (candidate indices 0,2,4,9) and four fixed probes (10,12,13,14) form independent branches from the initial state. Neither branch includes the other branch's observations. No action or hyperprior was selected on this final cohort. Finite-scale posterior endpoint mass should be compared with its prior value 0.4.

12. Temporal-prior follow-up

The v04 force-correlation failure motivated a separately frozen v06 follow-up. The original stress-test results remain in the tables. There are 32 fresh cases per geometry. True force correlation length is 0.10, and amplitude scale is sampled from the same five-value prior. The expanded estimator mixes lengths $\{0.10, 0.30, 0.60\}$ and the five scales, giving 46,875 spatial/hyperparameter components. Controls are amplitude-only uncertainty with nominal length 0.30, fixed nominal covariance, and an oracle given true covariance hyperparameters. The oracle does not receive true spatial material/support/load parameters. All methods receive identical original observations and the same separate four-view/four-probe branches.

Fixed true length 0.10 makes this a conditional stress test, not prior-averaged calibration. It is a follow-up to a recognized model error, not an untouched primary evaluation. Exact conditional Gaussian integration and full finite-mixture updates/intervals remain unchanged. An independent dense Gaussian identity audit verifies component indexing, normalized evidence and camera/probe updates to below 5×10^{-16}. Error, coverage and interval width are reported together, with every fresh case retained.

13. Continuous posterior calculation

Continuous inference uses the same five physical coordinates with a uniform bounded prior in those coordinates. Gaussian load amplitudes remain integrated exactly. Tempering selects the next likelihood exponent using a conditional effective-sample-size criterion. Systematic resampling and reflected random-walk Metropolis moves update particles, with bounded proposal adaptation between stages. Rejected or invalid states are not redrawn until valid. Stage-boundary checkpoints include RNG state and target/source fingerprints.

The main continuous cohort uses 32 fresh continuous-prior cases per geometry on the same mesh. Initial, recorded-active-final and recorded-fixed-final evidence are fitted with 1024 particles. The policies were selected by the finite dictionary, not by continuous SMC. A separate diagnostic repeats the first two finite-matched and off-grid-fine cases, both geometries and all three evidence stages, with two independent seeds and 1024/2048 particles. These checks evaluate sensitivity to numerical choices. The finite and continuous priors differ at the boundaries, so posterior disagreement is not solely quadrature error.

Targeted posthoc numerical check. Sensitivity flags for the block's matched case 1 after fixed measurements motivated two fresh-seed, 4096-particle repeats. The original 1024/2048 runs differ by up to 0.1104 in parameter marginal CDF and 0.0023 mm in target-mean RMS. Strong support/load and support/stiffness correlations are consistent with inefficient diagonal random-walk exploration; these summaries do not establish a separate missed mode. With the same four Metropolis moves per stage, the two larger runs differ by 0.0403 in CDF and 0.0003 mm in target-mean RMS, and agree within the original descriptive thresholds. Their final Metropolis acceptance rates are 12.9%/12.6%; resampled equal weights are not a convergence diagnostic. The maximum CDF difference between a 4096 run and a lower-count run is still 0.1006. All original flags and fits remain reported. This selected-case follow-up neither replaces the primary cohort nor certifies global convergence.

14. Excitation and forward-model qualification

A development study compared weak, intermediate and strong Gaussian supports before freezing the final cohort. Weak supports produced mostly rigid motion. The chosen 4–16 kN/m range raises median nonrigid/total RMS to 19.5% for block and 28.2% for step at the tested corner loads. Representative simultaneous 0.08 N loads give maximum principal strains 2.02%/2.61% and maximum rigid rotations 0.0173/0.0237 rad. These deterministic cases are not bounds on an untruncated Gaussian distribution.

A post-generation audit retains all 320 original surface-study truth histories. Maximum absolute principal strain is 4.30%, median case maximum 0.77%, and 95th percentile 1.90%. All cases lie below 5%. The smallest deformation-gradient determinant is 0.982. In the independent force-scale cohort, maximum strain is 8.18%, three of 128 cases exceed 5%, and none exceeds 10%. The smallest determinant is

Table 2. Posthoc temporal-prior follow-up, all 64 fresh cases. True correlation length 0.10; conditional stress coverage, not prior-averaged calibration. Force interval width is in N. Fixed views and probes branch independently.

Evidence	Model	Hidden mm	Reset mm	Material rel.	Pred. cov.	Force cov.	Width N
initial	scale-correlation	0.1456	0.0311	0.518	86.6	82.1	0.202
initial	amplitude-only	0.1550	0.0312	0.521	82.4	75.3	0.174
initial	fixed-nominal	0.1550	0.0322	0.506	77.2	68.4	0.135
initial	oracle	0.1379	0.0245	0.448	89.6	89.8	0.239
views	scale-correlation	0.1007	0.0304	0.506	88.0	83.9	0.177
views	amplitude-only	0.1170	0.0307	0.476	81.1	72.4	0.138
views	fixed-nominal	0.1170	0.0352	0.460	70.8	59.5	0.097
views	oracle	0.0972	0.0223	0.399	91.0	91.7	0.185
probes	scale-correlation	0.1450	0.0130	0.306	89.5	82.6	0.183
probes	amplitude-only	0.1541	0.0131	0.306	86.1	74.6	0.154
probes	fixed-nominal	0.1544	0.0157	0.331	80.3	68.8	0.127
probes	oracle	0.1373	0.0104	0.285	93.0	89.9	0.231

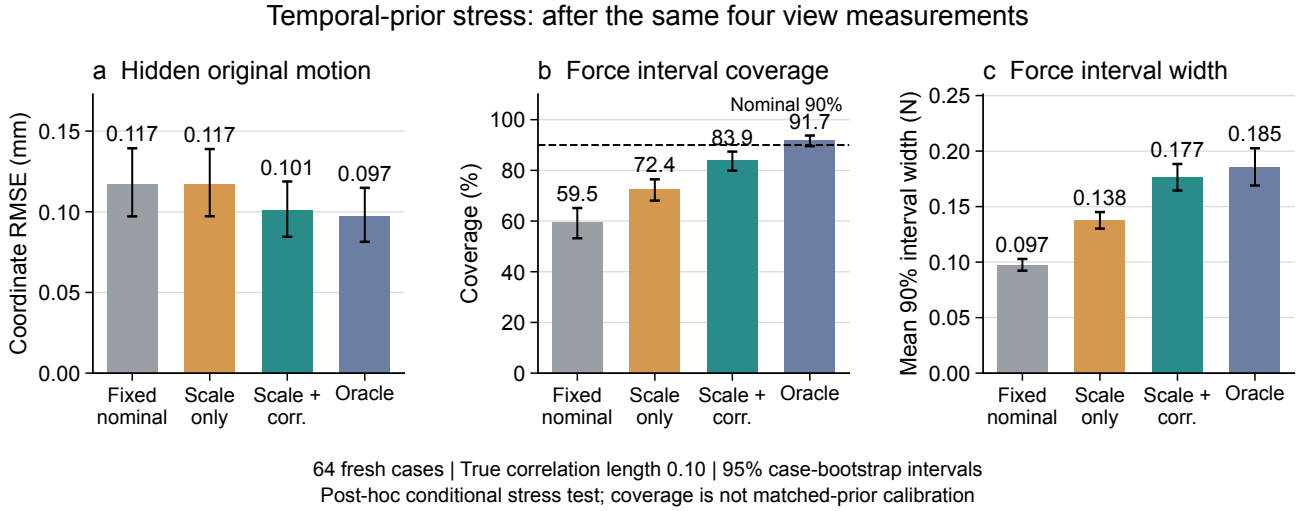


Figure 6. **Temporal uncertainty partially repairs a recognized failure.** Fresh 64-case conditional stress test after four fixed views. Error, coverage and width are shown together, with stratified case-bootstrap 95% intervals. The covariance oracle is given true amplitude scale and correlation length, not true material or force history.

0.958. These are checks on the declared linear generator, not validation against nonlinear elasticity.

A further development-only balanced-traction check was not promoted to replace the main action family. Opposing loads can reduce resultant force but introduce torque, while three-patch loads can cancel both. At equal total absolute actuator-force budget, a greater deformation fraction often coincided with a smaller absolute signal near the observation noise. Jacobian improvements depended on geometry and measurement location. This negative diagnostic guards against assuming that visually more deformable excitation always produces better identification.

Refinement checks use medium, fine and further-refined meshes at fixed physical parameters. The last $12 \rightarrow 16$ cell differences range from 0.008 to 0.068 mm, and primary $8 \rightarrow 16$ differences range from 0.027 to 0.209 mm. Some same-model inference errors are below discretization error. The fine-truth cohort exposes this limitation rather than treating same-mesh agreement as continuous-time or continuum physical accuracy.

15. Implementation, provenance and reproduction

For the quasistatic campaign, the priors, dictionary bounds, case counts, observation and target definitions, and seeds were fixed before final cohort execution. Case-level records retain every policy and every acquisition score. Continuous runs retain stage histories, invalid-state counts and independent seed/count comparisons. The tables give paired comparisons over the complete evaluation cohorts.

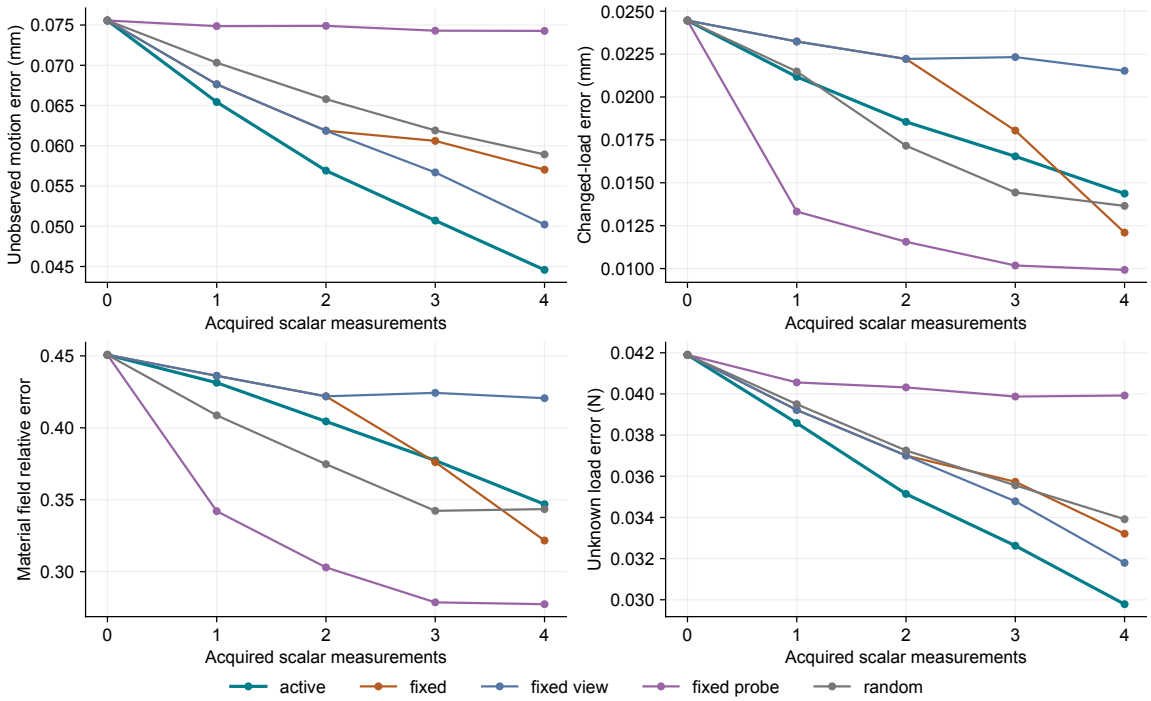


Figure 7. **All primary acquisition endpoints.** Complete matched-cohort trajectories for hidden motion, new loads, material and unknown-force error. Lower combined prediction error does not imply superiority on each component.

Numerical tests check rigid-mode strain, stiffness symmetry/positivity, force quadrature, gradients, CPU/GPU parity, affine interpolation, Gaussian marginal/conditional identities, covariance propagation, observation-order invariance, predictive-risk integration and strict target separation. The independent audit verifies that planners do not receive true parameters or unacquired measurements. Tests establish implementation identities and declared data flow, with numerical references.

Relation to dynamic RGB estimation. An earlier force-elimination experiment used a nonlinear Neo-Hookean trajectory and sparse rendered RGB. Its fifteen force coefficients were integrated exactly under a fixed full-rank velocity discrepancy model. A local Gaussian approximated the remaining trajectory/material posterior. It had no unknown spatial load footprint. Exact elimination therefore coexisted with approximate uncertainty and nonlinear renderer failures. The quasistatic reference isolates the conditional force calculation from image reconstruction and temporal integration.

16. Complete joint-field results

Each row summarizes a prespecified estimator or observation policy. Coordinate errors and marginal coverages are averaged by physical case. The first four tables separate matched and stress populations. Subsequent tables compare prior-scale models, continuous inference, posterior dependence, and the spatial distribution of traction.

Table 3. Matched finite. All errors are means of case-level metrics. Coverage is marginal at nominal 90%. Point/representation controls use initial evidence; acquisition rows use four additional scalars. Dashes indicate intervals not evaluated for that control.

Policy	Cases	Added	Hidden mm	New-load mm	Material rel.	Force N	Pred. cov.	Mat. cov.	Force cov.
Initial	128	0	0.0756	0.0245	0.451	0.042	91.1	90.5	90.3
active	128	4	0.0446	0.0144	0.347	0.030	92.0	90.7	91.6
fixed	128	4	0.0570	0.0121	0.322	0.033	89.2	87.7	90.8
fixed view	128	4	0.0502	0.0215	0.421	0.032	91.3	90.9	90.4
fixed probe	128	4	0.0743	0.0099	0.277	0.040	89.1	88.0	90.6
random	128	4	0.0589	0.0137	0.343	0.034	92.0	89.0	91.4
marginal map	128	0	0.0761	0.0274	0.553	0.043	–	–	–
joint map	128	0	0.0798	0.0607	0.517	0.050	–	–	–
uniform material	128	0	0.0762	0.0258	0.471	0.042	–	–	–
fixed interaction	128	0	0.2051	0.0495	0.666	0.134	–	–	–
independent moments	128	0	0.0759	0.0245	0.451	0.042	–	–	–

Table 4. Continuous coordinates. All errors are means of case-level metrics. Coverage is marginal at nominal 90%. Point/representation controls use initial evidence; acquisition rows use four additional scalars. Dashes indicate intervals not evaluated for that control.

Policy	Cases	Added	Hidden mm	New-load mm	Material rel.	Force N	Pred. cov.	Mat. cov.	Force cov.
Initial	64	0	0.0618	0.0246	0.465	0.046	90.2	96.9	90.0
active	64	4	0.0382	0.0172	0.393	0.033	86.4	93.9	91.1
fixed	64	4	0.0462	0.0163	0.397	0.037	85.9	93.3	87.9
fixed view	64	4	0.0422	0.0233	0.454	0.035	89.8	94.4	89.8
fixed probe	64	4	0.0617	0.0149	0.386	0.045	84.5	87.3	88.3
random	64	4	0.0488	0.0196	0.406	0.041	87.8	95.6	87.7
marginal map	64	0	0.0635	0.0334	0.752	0.048	–	–	–
joint map	64	0	0.0651	0.0665	0.552	0.054	–	–	–
uniform material	64	0	0.0615	0.0252	0.442	0.045	–	–	–
fixed interaction	64	0	0.1247	0.0368	0.588	0.084	–	–	–
independent moments	64	0	0.0623	0.0246	0.465	0.046	–	–	–

Table 5. Finer-mesh truth. All errors are means of case-level metrics. Coverage is marginal at nominal 90%. Point/representation controls use initial evidence; acquisition rows use four additional scalars. Dashes indicate intervals not evaluated for that control.

Policy	Cases	Added	Hidden mm	New-load mm	Material rel.	Force N	Pred. cov.	Mat. cov.	Force cov.
Initial	64	0	0.0732	0.0261	0.385	0.046	89.5	93.6	87.6
active	64	4	0.0473	0.0199	0.323	0.035	86.1	87.8	88.1
fixed	64	4	0.0574	0.0186	0.354	0.037	83.9	85.8	85.5
fixed view	64	4	0.0503	0.0253	0.367	0.034	88.6	90.8	89.0
fixed probe	64	4	0.0736	0.0195	0.352	0.045	81.9	79.8	85.3
random	64	4	0.0598	0.0220	0.400	0.038	84.6	85.5	86.6
marginal map	64	0	0.0752	0.0342	0.636	0.049	–	–	–
joint map	64	0	0.0746	0.0596	0.546	0.053	–	–	–
uniform material	64	0	0.0730	0.0278	0.381	0.045	–	–	–
fixed interaction	64	0	0.1558	0.0401	0.576	0.102	–	–	–
independent moments	64	0	0.0732	0.0261	0.385	0.046	–	–	–

Table 6. Force-correlation mismatch. All errors are means of case-level metrics. Coverage is marginal at nominal 90%. Point/representation controls use initial evidence; acquisition rows use four additional scalars. Dashes indicate intervals not evaluated for that control.

Policy	Cases	Added	Hidden mm	New-load mm	Material rel.	Force N	Pred. cov.	Mat. cov.	Force cov.
Initial	64	0	0.1257	0.0223	0.400	0.069	85.7	92.7	76.8
active	64	4	0.0931	0.0209	0.350	0.056	79.2	86.1	68.2
fixed	64	4	0.1011	0.0179	0.323	0.057	80.3	85.9	69.6
fixed view	64	4	0.0953	0.0284	0.380	0.055	79.6	80.0	67.7
fixed probe	64	4	0.1256	0.0129	0.315	0.068	83.7	90.3	75.8
random	64	4	0.1084	0.0199	0.303	0.059	82.0	84.8	72.2
marginal map	64	0	0.1261	0.0272	0.630	0.070	–	–	–
joint map	64	0	0.1279	0.0600	0.489	0.066	–	–	–
uniform material	64	0	0.1261	0.0228	0.422	0.069	–	–	–
fixed interaction	64	0	0.1936	0.0434	0.572	0.123	–	–	–
independent moments	64	0	0.1275	0.0223	0.400	0.069	–	–	–

Table 7. Fresh hierarchical force-scale cohort (128 cases). View/probe branches are independent. Oracle receives true prior scale only. Width is mean force marginal 90% interval width in N.

Evidence	Model	Hidden mm	New-load mm	Material rel.	Force N	Pred. cov.	Force cov.	Width N
initial	hierarchical	0.0939	0.0274	0.477	0.054	89.5	89.1	0.174
initial	fixed-nominal	0.0939	0.0294	0.478	0.055	84.2	81.2	0.140
initial	variance-matched	0.0941	0.0281	0.502	0.055	86.7	86.4	0.171
initial	oracle-scale	0.0936	0.0245	0.455	0.051	89.9	89.2	0.160
views	hierarchical	0.0626	0.0266	0.477	0.042	90.3	89.2	0.136
views	fixed-nominal	0.0635	0.0302	0.504	0.046	80.4	77.1	0.109
views	variance-matched	0.0632	0.0288	0.531	0.046	82.9	82.0	0.132
views	oracle-scale	0.0625	0.0234	0.453	0.040	90.0	89.0	0.121
probes	hierarchical	0.0929	0.0141	0.382	0.051	90.7	88.8	0.158
probes	fixed-nominal	0.0934	0.0160	0.404	0.052	85.5	81.9	0.134
probes	variance-matched	0.0932	0.0154	0.414	0.052	87.5	86.6	0.163
probes	oracle-scale	0.0928	0.0129	0.366	0.050	90.6	88.3	0.150

Table 8. Continuous-prior same-mesh cohort. 1024-particle SMC and finite dictionary receive identical evidence. Their priors differ; neither model selects new actions in this comparison.

Evidence	Model	Cases	Total mm	New-load mm	Material rel.	Pred. cov.	Force cov.
initial	continuous-smc	64	0.0481	0.0237	0.419	88.3	89.9
initial	finite-dictionary	64	0.0487	0.0246	0.465	90.2	90.0
active	continuous-smc	64	0.0294	0.0157	0.368	89.7	91.2
active	finite-dictionary	64	0.0309	0.0172	0.393	86.4	91.1
fixed	continuous-smc	64	0.0350	0.0152	0.362	88.6	88.2
fixed	finite-dictionary	64	0.0359	0.0163	0.397	85.9	87.9

Table 9. Joint dependence versus independent force moments on all 128 matched cases. Changed-load predictions and material marginals are identical by construction. Width averages 24 evaluated prediction coordinates. The control preserves exact marginal first/second moments but Gaussianizes the force marginal.

Evidence	Model	Hidden mm	Hidden cov.	Width mm
initial	full joint	0.0756	90.4	0.1171
initial	independent force moments	0.0759	93.1	0.1316
active	full joint	0.0446	91.7	0.0717
active	independent force moments	0.0449	95.4	0.0863

Table 10. Posthoc actual surface-traction diagnostic, all 128 matched cases. Area/time-weighted vector RMSE in Pa, averaged over cases; eight-time trapezoidal weighting. The independent control multiplies separate footprint/amplitude posterior means. It does not change fitted marginals.

Evidence	Estimator	Combined Pa	Top Pa	Side Pa
initial	joint	10.151	8.893	10.862
initial	independent	10.163	8.924	10.857
fixed	joint	8.038	7.420	8.184
fixed	independent	8.051	7.436	8.193
active	joint	7.180	6.940	6.824
active	independent	7.186	6.954	6.821

17. Finer-mesh inverse reference

A separately frozen numerical study compares an $8 \times 4 \times 4$ inverse model with a $24 \times 12 \times 12$ model on independent $32 \times 16 \times 16$ truth. Material contrast is volume-centered consistently across meshes. Stiffness factorizations are reused across force centers, but the physical model and parameter dictionary remain unchanged. Sixteen fresh cases per geometry use the same four-action budget and fixed target. This v03 diagnostic retains interior material sensors and is not a visible-surface camera experiment. It isolates discretization effects rather than pooling its errors with the v04 surface study. On 32 fresh high-resolution truths, initial coordinate RMSE is 0.0687 mm for the coarse inverse model and 0.0670 mm for the finer model. After their respective four adaptive measurements, errors are 0.0415/0.0403 mm and nominal 90% coverages are 76.7%/87.2%. Active actions may differ across inverse models; fixed schedules and potential observation noise remain paired. This comparison tests numerical model mismatch with interior sensors, not raw-video performance.

Table 11. Paired coarse/fine inverse reference on 32 independently generated 32-mesh truths. Both inverse models use the same 3125 parameter atoms. Sensors are interior material coordinates.

Inverse	Evidence	Total mm	Hidden mm	New-load mm	Material rel.	Pred. cov.
coarse8	initial	0.0687	0.0888	0.0325	0.457	78.1
fine24	initial	0.0670	0.0868	0.0306	0.456	85.2
coarse8	active	0.0415	0.0537	0.0205	0.378	76.7
fine24	active	0.0403	0.0521	0.0184	0.353	87.2
coarse8	fixed	0.0446	0.0581	0.0202	0.388	80.3
fine24	fixed	0.0417	0.0548	0.0175	0.361	88.2
coarse8	fixed-view	0.0502	0.0613	0.0297	0.424	77.2
fine24	fixed-view	0.0476	0.0581	0.0281	0.418	85.9
coarse8	fixed-probe	0.0615	0.0847	0.0169	0.390	71.9
fine24	fixed-probe	0.0599	0.0831	0.0120	0.299	86.6
coarse8	random	0.0534	0.0704	0.0206	0.390	75.8
fine24	random	0.0510	0.0681	0.0168	0.355	85.9

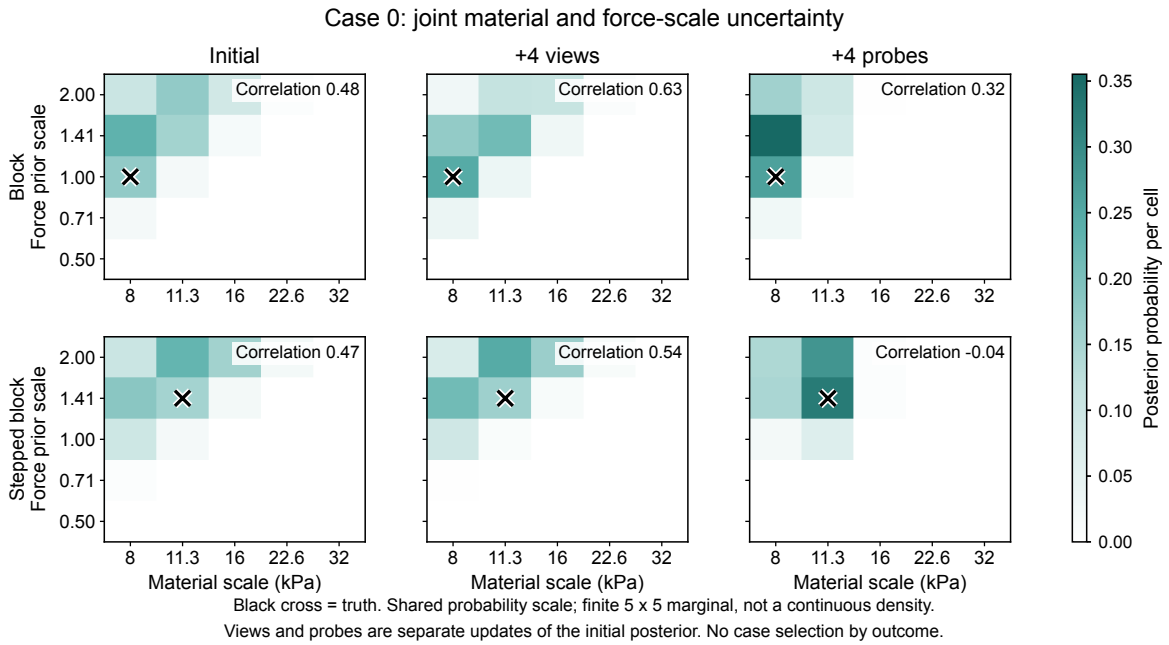


Figure 8. **Material and force scale after an observation.** In the first case for each geometry, the black cross marks the simulated values. The 5×5 grids show how additional views and probes update the marginal distribution over modulus and force scale.

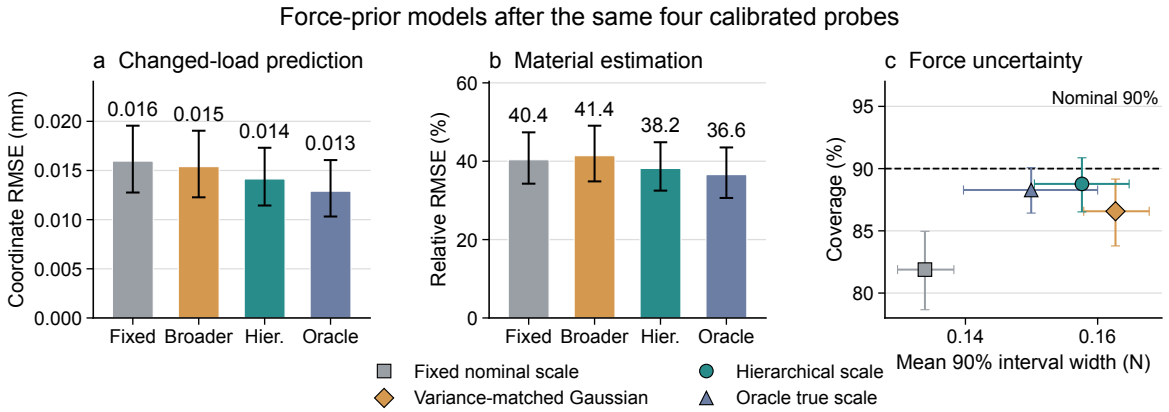


Figure 9. **Force-prior controls on 128 cases.** The broader Gaussian matches the hierarchy's prior variance. The oracle receives the true amplitude scale. Bars summarize performance after the same four probe measurements.